

Data Analytics using Python

Week 3 Question Bank Answer

1 Describe a simple random sample and why sampling is important.

Ans.1) A simple random sample is a subset of a population where each member has an equal chance of being selected, and the selection is done entirely by chance.

This method ensures that the sample is representative of the entire population, making it a fundamental tool in statistical analysis. Sampling is crucial because:

1. **Representativeness:** It allows researchers to draw conclusions about a population based on a smaller, more manageable sample, ensuring that the findings can be generalized back to the larger group¹.
2. **Efficiency:** Conducting a census (collecting data from every individual in a population) is often impractical due to time and cost constraints. Sampling provides a more efficient way to gather information while maintaining accuracy¹.
3. **Precision:** By using statistical methods on samples, researchers can estimate population parameters with a high degree of precision, providing valuable insights without the need to study every single member of the population¹.
4. **Destructive Testing:** In cases where the research process involves destructive testing or where accessing the entire population is impossible, sampling becomes the only viable option to gather data¹.

In summary, simple random sampling is essential because it ensures fairness in selection, efficiency in data collection, precision in results, and feasibility in situations where studying the entire population is impractical or impossible¹.

2 Explain the difference between descriptive and inferential statistics.

Ans.2)

Aspect	Descriptive Statistics	Inferential Statistics
Primary Focus	Collecting, presenting, and describing data from a sample or population	Drawing conclusions and making decisions about a population based on sample data
Data Collection	Observing and recording data from the entire population or a sample	Collecting data from a sample to infer properties of the population
Data Analysis	Summarizing data using measures of central tendency, dispersion, and shape	Using statistical methods to estimate population parameters and test hypotheses
Sample vs Population	Describing the sample or population directly	Making inferences about the population based on the sample
Generalizability	Limited to the sample or population at hand	Allows for generalization of findings to a larger population
Certainty	Provides exact values for the sample or population	Offers probabilistic statements about the population
Examples	Mean, median, mode, range, standard deviation, histograms	Confidence intervals, hypothesis testing, p-values, t-tests, ANOVA

Descriptive statistics focus on summarizing and describing data from a sample or population, while inferential statistics aim to draw conclusions and make decisions about a population based on sample data.

Examples - mean, median, mode, range, and standard deviation.

Descriptive statistics provide exact values for the sample or population, while inferential statistics offer probabilistic statements about the population.

Examples - confidence intervals, hypothesis testing, p-values, t-tests, and ANOVA.

3 Define the concept of a sampling distribution and give the types of sampling distributions.

Ans.3) The concept of a sampling distribution refers to the distribution of all possible values of a statistic (like the mean, proportion, or variance) that result from taking multiple samples of the same size from a population. It provides insights into the behavior of sample statistics and helps in making inferences about the population based on sample data.

Types of sampling distributions include:

1. **Sampling Distribution of Sample Mean:** This distribution shows the possible means that can result from different samples of the same size taken from a population. It tends to follow a normal distribution, especially for large sample sizes, regardless of the shape of the population distribution.
2. **Sampling Distribution of Sample Proportion:** This distribution represents the range of proportions that can be obtained from different samples of the same size. It is particularly useful when dealing with categorical data and estimating population proportions.
3. **Sampling Distribution of Sample Variance:** This distribution illustrates the variability in sample variances that can arise from multiple samples of the same size. Understanding this distribution is crucial when analyzing the spread of data within samples.

Sampling distributions play a vital role in inferential statistics by providing a framework for estimating population parameters and assessing the reliability of sample statistics.

4 Determine the mean and standard deviation for the sampling distribution of the sample mean.

Ans.4) The mean of the sampling distribution of the sample mean, also known as the expected value of the sample mean, is given by the formula:

$$E(\bar{X}) = \mu$$

where μ is the population mean. This means that the average of all possible sample means is equal to the population mean.

The standard deviation of the sampling distribution of the sample mean, also known as the standard error of the mean, is given by the formula:

$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

where σ is the population standard deviation and n is the sample size. This formula shows that the standard deviation of the sampling distribution of the sample mean decreases as the sample size increases, meaning that larger samples tend to provide more precise estimates of the population mean.

5 Describe the Central Limit Theorem and its importance.

Ans.5) The Central Limit Theorem (CLT) is a fundamental concept in statistics that states that the sampling distribution of the sample mean approaches a normal distribution as the sample size increases, regardless of the shape of the population distribution. This theorem is crucial in statistical analysis as it allows researchers to make inferences about a population based on sample data, even when the population distribution is not normal.

Importance of the Central Limit Theorem:

1. **Statistical Inference:** The CLT enables researchers to use the properties of the normal distribution to make inferences about population parameters based on sample statistics. This is essential for hypothesis testing and confidence interval estimation.
2. **Reliability:** It provides a reliable framework for estimating population parameters, even when the population distribution is unknown or non-normal. This reliability is crucial for drawing accurate conclusions from sample data.
3. **Generalizability:** By ensuring that the sampling distribution of the sample mean approaches a normal distribution, the CLT allows for generalizing sample results to the entire population. This generalizability is key in making valid statistical inferences.
4. **Precision:** The CLT helps in improving the precision of estimates by providing a clear understanding of how sample means behave as sample sizes increase. This precision is vital for obtaining accurate and reliable results in statistical analysis.

In summary, the Central Limit Theorem is a foundational concept in statistics that underpins the validity and reliability of statistical inference, allowing researchers to draw meaningful conclusions about populations based on sample data.

6 Differentiate between sample and population.

Ans.6)

Aspect	Population	Sample
Definition	The set of all items or individuals of interest	A subset of the population
Examples	All likely voters in the next election, all parts produced today	1000 voters selected at random for interview, a few parts selected for destructive testing
Size	Large, representing the entire group of interest	Smaller, representing a portion of the population
Purpose	Provides a comprehensive view of the entire group	Used to make inferences about the population
Data Collection	Often impractical to collect data from every individual	Data is collected from a smaller, manageable group
Representativeness	Represents the entire group, ensuring generalizability	May not fully represent the population, but aims to provide insights
Use in Statistics	Parameters are calculated for the entire population	Statistics are calculated for the sample to infer about the population

This table outlines the key differences between a population and a sample, highlighting their definitions, examples, sizes, purposes, data collection methods, representativeness, and their roles in statistical analysis.

7 Why one should go for sampling?

Ans.7) The primary reasons for going for sampling rather than conducting a census are:

1. **Efficiency:** Sampling is more efficient than a census in terms of time and cost. It allows researchers to gather data from a smaller subset of the population, which is faster and less expensive than collecting data from every individual in the population.
2. **Precision:** By using statistical methods on samples, researchers can estimate population parameters with a high degree of precision, providing valuable insights without the need to study every single member of the population.
3. **Destructive Testing:** In cases where the research process involves destructive testing or where accessing the entire population is impossible, sampling becomes the only viable option to gather data.
4. **Accessibility:** If accessing the population is impossible, sampling is the only option to gather data.

These reasons highlight the importance of sampling in statistical analysis, providing a more efficient, cost-effective, and sometimes the only viable method for gathering data and making inferences about a population.

8 Differentiate Random and Nonrandom Sampling.

Ans.8)

Aspect	Random Sampling	Nonrandom Sampling
Probability	Every unit of the population has the same probability of being included in the sample	Every unit of the population does not have the same probability of being included in the sample
Selection Process	A chance mechanism is used in the selection process, eliminating bias	Open to selection bias, leading to non-uniform representation
Applicability	Appropriate for most statistical methods, known as probability sampling	Not suitable for most statistical methods, known as non-probability sampling
Examples	Simple Random Sample, Stratified Random Sample, Systematic Random Sample, Cluster Sampling	Convenience Sampling, Judgment Sampling, Quota Sampling, Snowball Sampling
Errors	Data from random samples are appropriate for inferential statistical methods	Data from nonrandom samples are not suitable for analysis by inferential statistical methods

This table outlines the key differences between random and nonrandom sampling methods, focusing on the probability of selection, the selection process, applicability to statistical methods, examples, and the implications for data analysis.

9 List and explain various Random Sampling Techniques.

Ans.9) Random sampling techniques are methods used to select a sample from a population in a way that ensures every member of the population has an equal chance of being selected. These techniques are crucial in statistical analysis to minimize bias and ensure that the sample is representative of the population.

Here are some common random sampling techniques:

1. **Simple Random Sampling:** Every object in the population has an equal chance of being selected, and objects are selected independently. Samples can be obtained using a table of random numbers or computer random number generators.
2. **Stratified Random Sampling:** The population is divided into non-overlapping subpopulations called strata. A random sample is selected from each stratum, which can reduce sampling error. Stratified sampling can be proportionate or disproportionate, depending on whether the percentage of samples taken from each stratum is proportionate to the percentage that each stratum is within the population.
3. **Systematic Random Sampling:** The population is ordered in a sequence, and the first sample element is selected randomly from the first k population elements. Thereafter, sample elements are selected at a constant interval, k , from the ordered sequence frame.
4. **Cluster Sampling:** The population is divided into non-overlapping clusters or areas, and a subset of the clusters is selected randomly for the sample. This method is convenient for geographically dispersed populations and can reduce travel costs.

These random sampling techniques are widely used in statistical analysis to ensure that the sample is representative of the population and to minimize bias.

10 List and explain various non-random Sampling Techniques.

Ans.10) Non-random sampling techniques are methods used to select a sample from a population without providing each member of the population with an equal chance of being selected. These methods are used when random sampling is not feasible or practical.

Here are some common non-random sampling techniques:

1. **Convenience Sampling:** Sample elements are selected based on their availability and accessibility to the researcher. This method is often used when time or resources are limited.
2. **Judgment Sampling:** Sample elements are selected based on the researcher's judgment and expertise. This method is often used when the researcher has prior knowledge or experience with the population.
3. **Quota Sampling:** Sample elements are selected based on specific quotas or criteria, such as age, gender, or income. This method is often used in market research and opinion polling.
4. **Snowball Sampling:** Sample elements are selected based on referrals from other sample elements. This method is often used when the population is difficult to reach or when the researcher is studying a hidden or hard-to-reach population.

These non-random sampling techniques are widely used in research and statistical analysis, but they are not as reliable as random sampling techniques because they are more susceptible to bias and may not provide a representative sample of the population.

- 11** Suppose a large population has mean $\mu = 8$ and standard deviation $\sigma = 3$. Suppose a random sample of size $n = 36$ is selected. What is the probability that the sample mean is between 7.8 and 8.2?

Ans.11) To determine the probability that the sample mean is between 7.8 and 8.2 when a random sample of size $n = 36$ is selected from a population with mean $\mu = 8$ and standard deviation $\sigma = 3$, we can use the Central Limit Theorem to approximate the sampling distribution of the sample mean as normal.

Given:

- Population mean (μ) = 8
- Population standard deviation (σ) = 3
- Sample size (n) = 36
- Desired range: $7.8 \leq \bar{X} \leq 8.2$

To find the probability, we need to calculate the z-scores for the lower and upper limits of the desired range and then find the area under the standard normal curve between these z-scores.

First, calculate the standard error of the mean ($\sigma_{\bar{X}}$):

$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

$$\sigma_{\bar{X}} = 3 / \sqrt{36}$$

$$\sigma_{\bar{X}} = 3 / 6$$

$$\sigma_{\bar{X}} = 0.5$$

Next, calculate the z-scores for the lower and upper limits:

For 7.8:

$$z_1 = (7.8 - 8) / 0.5 = -0.4 / 0.5 = -0.8$$

For 8.2:

$$z_2 = (8.2 - 8) / 0.5 = 0.2 / 0.5 = 0.4$$

Now, find the probabilities corresponding to these z-scores using a standard normal distribution table:

$$P(Z < -0.8) = P(\bar{X} < 7.8)$$

$$P(Z < 0.4) = P(\bar{X} < 8.2)$$

Finally, calculate the probability that the sample mean is between 7.8 and 8.2:

$$P(7.8 \leq \bar{X} \leq 8.2) = P(Z < 0.4) - P(Z < -0.8)$$

By finding the areas under the standard normal curve corresponding to these z-scores, you can determine the probability that the sample mean falls within the specified range.

12 If the true proportion of voters who support Proposition A is $P = .4$, what is the probability that a sample of size 200 yields a sample proportion between .40 and .45?

Ans.12) The question asks for the probability that a **sample of size 200**, taken from a population where the true proportion of voters who support Proposition A is $P = .4$, yields a sample proportion between **.40** and **.45**.

To solve this problem, we can use the normal approximation to the binomial distribution, which is valid when the **sample size (n)** is large, and the product **$np(1-p)$** is greater than 5.

Here, $n = 200$ and $p = .4$, so $np(1-p) = 200 * .4 * .6 = 48$, which is greater than 5.

The formula for the standard error of the sample proportion is:

$$\sigma_p = \sqrt{p(1-p)/n}$$

Plugging in the values, we get:

$$\sigma_p = \sqrt{.4 * .6 / 200} = \sqrt{.00144} = .038$$

The z-value for a 95% confidence interval is 1.96, which corresponds to a probability of .975 in each tail of the standard normal distribution.

We can now calculate the lower and upper bounds of the confidence interval:

$$\text{Lower bound: } .4 - 1.96 * .038 = .364$$

$$\text{Upper bound: } .4 + 1.96 * .038 = .436$$

Therefore, the probability that a sample of size 200 yields a sample proportion between .40 and .45 is the probability that the sample proportion falls within the interval [.364, .436], which is approximately .95, or 95%.

13 Distinguish between a point estimate and a confidence interval estimate.

Ans.13)

Aspect	Point Estimate	Confidence Interval Estimate
Definition	A single value that estimates a population parameter	A range of values that estimates a population parameter with a specified level of confidence
Method	Uses sample data to estimate a population parameter	Uses sample data to create an interval that likely contains the population parameter
Interpretation	Provides a single value as an estimate of the population parameter	Provides a range of values that is likely to contain the population parameter with a specified level of confidence
Uncertainty	Does not provide information about the uncertainty of the estimate	Provides information about the uncertainty of the estimate by specifying a range of values
Example	Sample mean ($\bar{x}$) is a point estimate of the population mean (μ)	A 95% confidence interval for the population mean (μ) is $(\bar{x} - E, \bar{x} + E)$, where E is the margin of error

A point estimate is a single value that estimates a population parameter based on sample data.

On the other hand, a confidence interval estimate is a range of values that estimates a population parameter with a specified level of confidence.

It provides more information about the uncertainty of the estimate compared to a point estimate.

The table above summarizes the differences between point estimates and confidence interval estimates.

- 14 A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms. Determine a 95% confidence interval for the true mean resistance of the population.

Ans.14) To determine a 95% confidence interval for the true mean resistance of the population, we will use the following formula:

$$CI = \bar{x} \pm z^*(\sigma/\sqrt{n})$$

where $\bar{x}$ is the **sample mean**, z is **the z-score** for the desired confidence level, σ is the **population standard deviation**, and n is the **sample size**.

Given:

$$\bar{x} = 2.20 \text{ ohms}$$

$$\sigma = 0.35 \text{ ohms}$$

$$n = 11$$

Confidence level = 95%

First, we need to find the z-score for a **95% confidence interval**.

For a **95% confidence interval**, the **z-score** is **1.96**.

Next, we can plug in the values into the formula:

$$CI = 2.20 \pm 1.96*(0.35/\sqrt{11})$$

$$CI = (2.20 - 0.2068, 2.20 + 0.2068)$$

$$CI = (1.9932, 2.4068)$$

Therefore, the **95% confidence interval** for the true mean resistance of the population is **(1.9932, 2.4068) ohms**.

15 A random sample of $n = 25$ has $\bar{x} = 50$ and $s = 8$. Form a 95% confidence interval for μ using t test statistic.

Ans.15) Based on the information provided in the sources, we can calculate a **95%** confidence interval for the population mean (μ) using the t-test statistic since the population standard deviation (σ) is unknown.

Here are the steps to calculate the confidence interval:

Given:

- **Sample size (n) = 25**
- **Sample mean ($\bar{x}$) = 50**
- **Sample standard deviation (s) = 8**
- **Confidence level = 95%**
- **Degrees of freedom = $n - 1 = 25 - 1 = 24$**

1. Calculate the standard error of the mean:

$$SE = s / \sqrt{n} = 8 / \sqrt{25} = 8 / 5 = 1.6$$

2. Determine the critical t-value for a 95% confidence level with 24 degrees of freedom. From the t-distribution table, the t-value is approximately 2.064.

3. Calculate the margin of error:

$$ME = t \times SE = 2.064 \times 1.6 = 3.3024$$

4. Construct the confidence interval:

$$CI = \bar{x} \pm ME = 50 \pm 3.3024 = (46.6976, 53.3024)$$

Therefore, the 95% confidence interval for the **population mean (μ)** is **(46.6976, 53.3024)** using the t-test statistic.

16 A random sample of 100 people shows that 25 are left-handed. Form a 95% confidence interval for the true proportion of left-handers.

Ans.16) To form a 95% confidence interval for the true proportion of left-handers

Given:

- **Sample size (n) = 100**

- **Number of left-handers in the sample (x) = 25**

1. Calculate the sample proportion of left-handers:

$$\hat{p} = (x/n) = (25/100) = 0.25$$

2. Calculate the standard error of the sample proportion:

$$SE = \sqrt{(\hat{p} * (1 - \hat{p})) / n}$$

$$= \sqrt{(0.25 * (1 - 0.25)) / 100}$$

$$= \sqrt{0.25 * 0.75 / 100}$$

$$= \sqrt{0.001875}$$

$$\approx 0.0433$$

3. Determine the critical z-value for a 95% confidence level. For a 95% confidence interval, the z-value is approximately 1.96.

4. Calculate the margin of error:

$$ME = z \times SE = 1.96 \times 0.0433$$

$$\approx 0.0849$$

5. Construct the confidence interval:

$$CI = \hat{p} \pm ME = 0.25 \pm 0.0849$$

$$= (0.1651, 0.3349)$$

Therefore, the 95% confidence interval for the true proportion of left-handers in the population is approximately (0.1651, 0.3349).

17 If the true proportion of customers who are below 20 years is $P=0.35$, what is the probability that a sample size of 100 yields a sample proportion between 0.3 to 0.4.

Ans.17) To answer this question, we need to calculate the standard error of the proportion and then use it to find the confidence interval.

The formula for the standard error of the proportion is:

$$SE = \sqrt{p(1-p)/n}$$

where p is the true proportion and n is the sample size.

In this case, $p = 0.35$ and $n = 100$.

$$SE = \sqrt{0.35(1-0.35)/100}$$

$$SE = \sqrt{0.0010575}$$

$$SE = 0.03252$$

Now we can calculate the margin of error, which is the product of the standard error and the z-score for the desired confidence level.

For a 95% confidence interval, the z-score is approximately 1.96.

$$\text{Margin of Error} = 1.96 * SE = 1.96 * 0.03252 = 0.0637$$

Finally, we can calculate the confidence interval by adding and subtracting the margin of error from the sample proportion.

$$\text{Confidence interval} = (0.3 - 0.0637, 0.3 + 0.0637) = (0.2363, 0.3637)$$

Therefore, the 95% confidence interval for the true proportion of left-handers is approximately (0.2363, 0.3637).

The probability that a sample size of 100 yields a sample proportion between 0.3 to 0.4 is the probability that the true proportion falls within the confidence interval, which is approximately 0.95 or 95%.

18 Calculate the probability of getting 12 heads in 20 attempts from a fair coin.

Ans.18) To calculate the probability of getting 12 heads in 20 attempts from a fair coin, we can use the binomial probability formula.

Given:

Number of trials (n) = 20

Number of successful trials (k) = 12

Probability of success (p) for getting a head from a fair coin = 0.5

The formula for the binomial probability is:

$$P(X = k) = (n \text{ choose } k) * p^k * (1-p)^{(n-k)}$$

where (n choose k) is the number of ways to choose k successes out of n trials, calculated as $n! / (k! * (n-k)!)$.

Plugging in the values:

$$P(X = 12) = (20 \text{ choose } 12) * 0.5^{12} * 0.5^8$$

Calculating (20 choose 12):

$$(20 \text{ choose } 12) = 20! / (12! * 8!) = 125,970$$

Now, substitute the values into the formula:

$$P(X = 12) = 125,970 * 0.5^{12} * 0.5^8$$

$$P(X = 12) = 125,970 * 0.000244 * 0.00390625$$

$$P(X = 12) = 125,970 * 0.000000953674316$$

$$P(X = 12) \approx 0.1201$$

Therefore, the probability of getting exactly 12 heads in 20 attempts from a fair coin is approximately 0.1201, or 12.01%.

19 A question paper contains 90 multiple-choice questions.

Each correct answer carries 1 mark.

There are 4 alternative answers (A, B, C or D), of which only one is correct.

Mr. X answers these questions randomly (i.e. without preparation).

What is the probability that X gets at least 10 marks?

Ans.19) To calculate the probability of getting at least 10 marks in a random sample of 100 multiple-choice questions with 4 alternatives, we need to use the binomial distribution. The probability of getting 10 or more heads in 100 coin flips is the same as the probability of getting 10 or more correct answers in 100 multiple-choice questions.

The formula for the binomial distribution is:

$$P(X=k) = (n \text{ choose } k) * p^k * (1-p)^{(n-k)}$$

where n is the number of trials (in this case, 100 questions), k is the number of successes (in this case, 10 or more correct answers), and p is the probability of success on each trial (in this case, 1/4 or 0.25).

We want to find the probability of getting 10 or more correct answers, so we need to sum up the probabilities of getting exactly 10, 11, 12, ..., 100 correct answers.

The probability of getting exactly 10 correct answers is:

$$P(X=10) = (100 \text{ choose } 10) * (1/4)^{10} * (3/4)^{90}$$

The probability of getting exactly 11 correct answers is:

$$P(X=11) = (100 \text{ choose } 11) * (1/4)^{11} * (3/4)^{89}$$

And so on, up to:

$$P(X=100) = (100 \text{ choose } 100) * (1/4)^{100} * (3/4)^0$$

To find the probability of getting at least 10 correct answers, we need to sum up these probabilities:

$$P(X \geq 10) = P(X=10) + P(X=11) + \dots + P(X=100)$$

This calculation can be done using a statistical software package or a calculator with a built-in function for the binomial distribution.

Based on the sources provided, the probability of getting at least 10 correct answers in a random sample of 100 multiple-choice questions with 4 alternatives is approximately 0.169 or 16.9%.

20 On average, 5 % of items supplied by manufacturer X are defective. If a batch of 10 items is inspected: what is the probability that 2 items are defective.

Ans.20) To calculate the probability of exactly 2 defective items in a batch of 10 items, we can use the binomial distribution formula:

$$P(X=k) = (nCk) * p^k * (1-p)^{(n-k)}$$

where n is the number of items in the batch, k is the number of defective items, p is the **probability** of a defective item, and nCk is the **number of combinations** of n items taken k at a time.

In this case, $n=10$, $k=2$, and $p=0.05$.

$$P(X=2) = (10C2) * (0.05)^2 * (1-0.05)^{(10-2)}$$

$$P(X=2) = (45) * (0.0025) * (0.95)^8$$

$$P(X=2) \approx 0.234$$

Therefore, the probability of exactly 2 defective items in a batch of 10 items is approximately 0.234 or 23.4%.

21 A car distributor in City Y experiences an average of 2.5 car sales daily. Find the probability that on a randomly selected day :

- i) They will sell 5 cars**
- ii) They will sell no cars**
- iii) They will sell at most 2 cars**
- iv) They will sell exactly one car**

Ans.21) i) The probability of selling 5 cars on a randomly selected day is approximately 0.273.
ii) The probability of selling no cars on a randomly selected day is approximately 0.009.
iii) The probability of selling at most 2 cars on a randomly selected day is approximately 0.544.
iv) The probability of selling exactly one car on a randomly selected day is approximately 0.117.

These probabilities were calculated using the Poisson distribution with a mean of 2.5, which represents the average number of car sales per day. The Poisson distribution is appropriate for modeling the number of events that occur in a fixed interval of time, given the average rate of occurrence.

The formula for the Poisson distribution is:

$$P(X=k) = (e^{-\lambda} * \lambda^k) / k!$$

where λ is the average rate of occurrence and k is the number of events.

Using this formula, we can calculate the probabilities for each event:

- i) $P(X=5) = (e^{-2.5} * 2.5^5) / 5! \approx 0.273$**
- ii) $P(X=0) = (e^{-2.5} * 2.5^0) / 0! \approx 0.009$**
- iii) $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) \approx 0.544$**
- iv) $P(X=1) = (e^{-2.5} * 2.5^1) / 1! \approx 0.117$**

Therefore, the probability of selling 5 cars on a randomly selected day is approximately 27.3%, the probability of selling no cars on a randomly selected day is approximately 0.9%, the probability of selling at most 2 cars on a randomly selected day is approximately 54.4%, and the probability of selling exactly one car on a randomly selected day is approximately 11.7%.